

Regular expression

Assuming that alphabet set Σ is $\{0,1\}$, lets check out the following examples of Regular Expression to Regular Language.

Regular Expression	Regular Language	Language Set
$(0+1)$	$\{w \mid \text{length of } w \text{ is one}\}$ <i>It can also be written as $(0U1)$ or $(0 1)$ or Σ</i>	$\{0, 1\}$
$(0+1)(0+1)$	$\{w \mid \text{length of } w \text{ is two}\}$	$\{00, 01, 10, 11\}$
$(0+1)^*$	$\{w \mid \text{length of } w \text{ is 0 or more}\}$	$\{\epsilon, 0, 1, 00, 01, 10, 11, 000, 001, \dots\}$
$(0+1)^+$	$\{w \mid \text{length of } w \text{ is 1 or more}\}$	$\{0, 1, 00, 01, 10, 11, 000, \dots\}$
0^*	$\{w \mid w \text{ is any number of 0's}\}$	$\{\epsilon, 0, 00, 000, 0000, \dots\}$
$(01)^*$	$\{w \mid w \text{ is any number of 01 substring}\}$	$\{\epsilon, 01, 0101, 010101, \dots\}$
$0(0+1)^*$	$\{w \mid w \text{ starts with 0}\}$	$\{0, 00, 01, 010, 011, \dots\}$
0^*10^*	$\{w \mid w \text{ contains a single 1}\}$	$\{1, 01, 010, 000100, \dots\}$
$\Sigma^*1\Sigma^*$	$\{w \mid w \text{ has at least one 1}\}$	$\{1, 11, 01, 10, 010, 110, 111, \dots\}$
$\Sigma^*001\Sigma^*$	$\{w \mid w \text{ has a substring 001}\}$	$\{001, 0001, 110010110, \dots\}$
$(\Sigma\Sigma)^*$	$\{w \mid w \text{ is a string of even length}\}$	$\{\epsilon, 00, 01, 10, 11, 1100, 0010, \dots\}$
$(\Sigma\Sigma\Sigma)^*$	$\{w \mid \text{the length of } w \text{ is a multiple of 3}\}$	$\{\epsilon, 011, 110100, \dots\}$
$0\Sigma^*0 \cup 1\Sigma^*1 \cup 0 \cup 1$	$\{w \mid w \text{ starts and ends with same character}\}$	$\{0, 1, 010, 111, 1010101, \dots\}$

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Examples of Regular Language to Regular Expression

▼ $L_0 = \{w \in \{0, 1\}^* \mid w \text{ contains } 101 \text{ as a substring}\}$

$$\Sigma^*101\Sigma^*$$

▼ $L_1 = \{w \in \{0, 1\}^* \mid w \text{ contains } 101 \text{ as a subsequence}\}$

$$\Sigma^*1\Sigma^*0\Sigma^*1\Sigma^*$$

▼ $L_2 = \{w \in \{0, 1\}^* \mid w \text{ starts with } 101\}$

$$101\Sigma^*$$

▼ $L_3 = \{w \in \{0, 1\}^* \mid w \text{ ends with } 101\}$

$$\Sigma^*101$$

▼ $L_4 = \{w \in \{0, 1\}^* \mid w \text{ contains } 00 \text{ or } 11\}$

$$\Sigma^*(00+11)\Sigma^*$$

▼ $L_5 = \{w \in \{0, 1\}^* \mid w \text{ contains exactly two } 1s\}$

$$0^*10^*10^*$$

▼ $L_6 = \{w \in \{0, 1\}^* \mid w \text{ contains at least two } 1s\}$

$$\Sigma^*1\Sigma^*1\Sigma^*$$

▼ $L_7 = \{w \in \{0, 1\}^* \mid w \text{ contains at most two } 1s.\}$

$$0^*(1+\varepsilon)0^*(1+\varepsilon)0^*$$

▼ $L_8 = \{w \in \{0, 1\}^* \mid \text{length of } w \text{ is even or divisible by } 2\}$

$$(\Sigma\Sigma)^*$$

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▼ $L_9 = \{w \in \{0, 1\}^* \mid \text{length of } w \text{ is odd}\}$

$$\Sigma(\Sigma\Sigma)^*$$

▼ $L_{10} = \{w \in \{0, 1\}^* \mid \text{length of } w \text{ is divisible by 3}\}$

$$(\Sigma\Sigma\Sigma)^*$$

▼ $L_{11} = \{w \in \{0, 1\}^* \mid \text{length of } w \text{ is not divisible by 3}\}$

$$(\Sigma\Sigma\Sigma)^*\Sigma + (\Sigma\Sigma\Sigma)^*\Sigma\Sigma$$

▼ $L_{12} = \{w \in \{0, 1\}^* \mid \text{number of 1's in } w \text{ is divisible by 3}\}$

$$(0^*10^*10^*10^*)^*$$

▼ $L_{13} = \{w \in \{0, 1\}^* \mid w \text{ starts and ends with different symbols}\}$

$$0\Sigma^*1 + 1\Sigma^*0$$

▼ $L_{14} = \{w \in \{0, 1\}^* \mid w \text{ starts and ends with the same symbols}\}$

$$0\Sigma^*0 + 1\Sigma^*1 + 0 + 1$$

▼ $L_{15} = \{w \in \{0, 1\}^* \mid w \text{ doesn't end with } 01\}$

$$\Sigma^*(00 + 10 + 11) + 0 + 1 + \varepsilon \quad \text{OR} \quad \Sigma^*(0 + 11) + 1 + \varepsilon$$

▼ $L_{16} = \{w \in \{0, 1\}^* \mid 0\text{s and } 1\text{s alternate in } w\}$

$$(01)^*(0+\varepsilon) + (10)^*(1+\varepsilon) \quad \text{OR} \quad (1+\varepsilon)(01)^*(0+\varepsilon)$$

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▼ $L_{17} = \{w \in \{0, 1\}^* \mid w \text{ doesn't contain } 00\}$

$$1^*(01^+)^*(0+\varepsilon)$$

▼ $L_{18} = \{w \in \{0, 1\}^* \mid w \text{ doesn't contain } 10\}$

$$0^*1^*$$

▼ $L_{19} = \{w \in \{0, 1\}^* \mid w \text{ doesn't contain } 00 \text{ and } 11\}$

$$(1+\varepsilon)(01)^*(0+\varepsilon)$$

Same as L_{16}

▼ $L_{20} = \{w \in \{0, 1\}^* \mid w \text{ contains } 0 \text{ in every third position}\}$

$$(\Sigma\Sigma 0)^*(\varepsilon + \Sigma + \Sigma\Sigma)$$